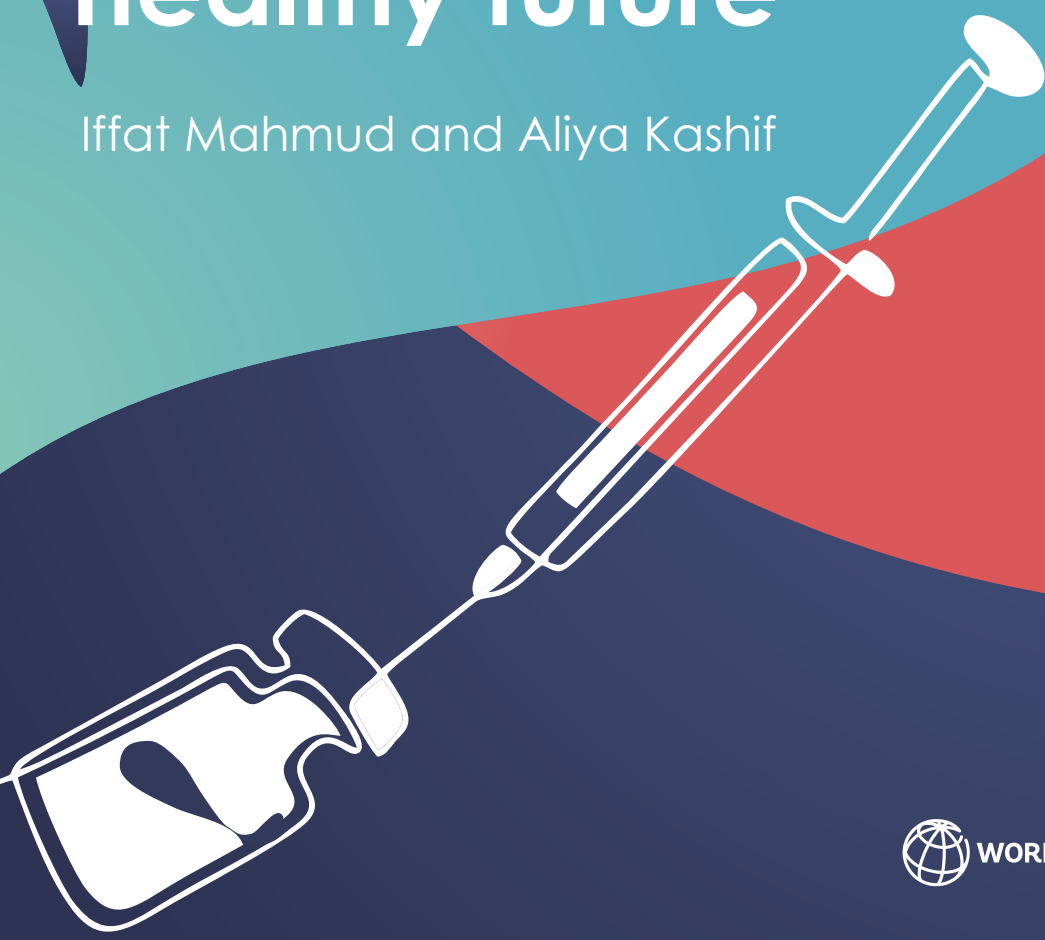




Immunization for Pakistan's healthy future

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Acronyms

BCG	Bacille Calmette-Guérin
CCT	Conditional Cash Transfers
COVID-19	Novel coronavirus disease
DTP	Diphtheria, Tetanus and Pertussis
EPI	Expanded Program on Immunization
FY	Fiscal Year
GDP	Gross Domestic Product
GNI	Gross National Income
Hep B	Hepatitis B
Hib	Haemophilus influenza type b
HNP	Health, Nutrition, and Population
IPV	Inactivated Polio Vaccine
KP	Khyber Pakhtunkhwa
NISP	National Immunization Support Project
OPV	Oral Polio Vaccine
PBS	Pakistan Bureau of Statistics
PDHS	Pakistan Demographic and Health Survey
PPP	Purchasing Power Parity
TPVICS	Third-Party Verification Immunization Coverage Survey
UNICEF	United Nations Children's Fund
VPD	Vaccine Preventable Disease
WHO	World Health Organization

Executive Summary

Pakistan has made impressive progress in immunization coverage over the past 30 years with a 117.6% increase in the proportion of children aged 12–23 months fully immunized between 1990 and 2020. With improvements in the vaccination rates, Pakistan was able to reduce the incidence of two vaccine preventable diseases (VPDs) – diphtheria by 94% and measles by 88% over the same period. Despite successive achievements, disparities remain in terms of coverage by vaccine type, socioeconomic background, and demographics.

Immunization remains a high priority for the Government of Pakistan and an important element of progress toward universal health coverage and resilient health systems. The government initiated the National Immunization Support Project (NISP) in 2016 with financing from partners including the World Bank, the United States Agency for International Development, Gavi the Vaccine Alliance, and the Bill and Melinda Gates Foundation. NISP is a five-year project, scheduled to close in June 2022, which supports and incentivizes the implementation of the government's Expanded Program on Immunization (EPI). NISP has been instrumental in reducing childhood morbidity and mortality from VPDs.

Despite significant efforts by the Government and its partners, the immunization coverage is yet to reach the expected benchmarks. Worldwide, Pakistan is one of the two countries where poliomyelitis is still endemic. With respect to coverage of vaccination for diphtheria, tetanus and pertussis/whooping cough, and measles, Pakistan had the second lowest coverage in the South Asia region in 2020. In terms of incidence of VPDs, Pakistan had the second highest incidence of both diphtheria and polio in South Asia in 2020. In comparison to the countries in Eastern and Southern Asia with similar income levels as Pakistan, immunization coverage is relatively low.

There is a need to continue to strengthen Pakistan's EPI to improve the health of its children. Further enhancing health systems through an expansion of vaccination sites, among others, and prioritizing lagging areas to maximize immunization coverage in underperforming provinces and districts, are two important. In addition, ensuring sustainable financing for EPI, fostering public-private partnership, and exploring innovative approaches for improving vaccine uptake and ensuring readiness of the supply side, can help improve immunization coverage.

1. Background



Introduction

Pakistan is the fifth most populous country in the world, with a projected population of 232 million in 2021, of whom 51 million are women of childbearing age and 33 million children under five years of age. The gross national income per capita was US\$1,410 in 2019. Periodic macroeconomic crises and a low human capital basis have constrained the country's growth prospects. Between fiscal years (FY) 2019 and 2020, annual economic growth, measured using gross domestic product (GDP), averaged around 4.3% in Pakistan, below the South Asian annual average of 6.2%. According to the World Bank Human Capital Index,¹ if no improvements in health and education service delivery take place, a Pakistani child born today can expect to be only 40% as productive as s/he otherwise could be by age 18.

Pakistan has witnessed some improvements in health, nutrition, and population (HNP) outcomes over the years; significant challenges however remain, especially given the progress made by countries of comparable wealth. Pakistan was not able to achieve any of the 2015 HNP-related millennium development goals; and, at current trends, is unlikely to meet most of the HNP-related Sustainable Development Goals by 2030. Goal 3 of the SDGs is to: “ensure healthy lives and promote wellbeing for all at all ages”. Two of the targets of this goal are, to end preventable deaths of newborns and children under five years of age, with all countries aiming to reduce neonatal mortality—death within the first 28 days of birth—to no more than 12 per 1,000 live births, and under-five mortality (the probability of a child dying before their fifth birthday) to no more than 25 per 1,000 live births by 2030. Pakistan reduced its neonatal mortality rate from 53.3 per 1,000 live births in 1990 to 41 deaths per 1,000 live births in 2019, but this rate was still the second highest in the world.² Similarly, the under-five child mortality rate) declined from 120.4 per 1,000 live births in 1990 to 74 per 1,000 live births in 2019.³

Childhood immunization against vaccine preventable diseases (VPD) presents a highly cost-effective intervention by delivering significant reductions in morbidity and mortality from inexpensive and standardized interventions. It remains one of the most fundamental competencies of public health programs. The Expanded Program on Immunization (EPI) has existed in Pakistan since 1978. In line with international standards, EPI in Pakistan aims to immunize all children against VPDs that include childhood tuberculosis, poliomyelitis, diphtheria, tetanus, pertussis, hepatitis B, haemophilus influenza type b (Hib) pneumonia and meningitis, measles, and diarrhea due to rotavirus.⁴

Immunization is a high priority for the Government of Pakistan and an important part of making progress to universal health coverage and building resilient health systems. The government initiated the National Immunization Support Project (NISP) in 2016 with financing from partners including the World Bank, the United States Agency for International Development, Gavi the Vaccine Alliance, and the Bill and Melinda Gates Foundation. NISP is a five-year project, scheduled to close in June 2022, that aims to increase equitable coverage of services for immunization against VPDs, including poliomyelitis, for children aged 0–23 months in Pakistan. The project supports and incentivizes the implementation of the government's EPI program, which has contributed to significantly decreasing childhood morbidity and mortality from VPDs. Despite significant efforts by the government and its partners, Pakistan's immunization indicators have yet to reach the expected benchmarks. Pakistan is one of the two countries in the world, where poliomyelitis is still endemic.⁵ The country ranks third in terms of highest number of unimmunized children (1.2 million) in the world, after Nigeria (approximately 4 million) and India (2.9 million).⁶

¹ https://databank.worldbank.org/data/download/hci/HCI_2pager_PAK.pdf

² data.worldbank.org

³ Pakistan Demographic and Health Survey 2017-18. 2019. National Survey, Islamabad: National Institute of Population Studies (NIPS) and ICF.

⁴ <http://www.emro.who.int/pak/programmes/expanded-programme-on-immunization.html>

⁵ Global Polio Eradication Initiative. 2021. <https://polioeradication.org/where-we-work/polio-endemic-countries/>

⁶ UNICEF. 2021. <https://www.unicef.org/pakistan/press-releases/over-20-million-children-worldwide-missed-out-measles-vaccine-annually-past-8-years>

Objectives of the report

This report documents the progress made by Pakistan in immunizing its children between 1990 and 2020. This is based on analysis of primary data collected through a nationwide survey, the Third-Party Verification Immunization Coverage Survey (TPVICS) 2021, as well as data from existing sources including the Pakistan Demographic and Health Survey (PDHS). The next chapter documents the significant progress made in immunization coverage over time, followed by a regional comparison of immunization status in South Asia to highlight where Pakistan stands. The report concludes with some programmatic recommendations.

Data sources

This report is based on secondary analysis of survey data including the TPVICS 2021 and PDHS. Demographic Health Surveys are standard internationally accepted surveys administered in nearly 90 countries across the world to collect information on selected health, nutrition, and population outcomes.

The cross-sectional TPVICS survey was carried out on children aged 12–23 months with a sample larger than that used for PDHS 2018. TPVICS covered almost 98,000 children, whereas PDHS 2018 covered 2,463 children of 12–23 months. The Pakistan Bureau of Statistics (PBS) was consulted for finalizing the sampling frame of the survey. Data was collected over the period from September 2020, starting with Punjab, and ending in January 2021 with Khyber Pakhtunkhwa (KP). The survey covered 97,760 children across the country, of whom 53,063 were male and 44,697 female. For the purposes of TPVICS, a fully immunized child is defined as one who has completed all immunizations starting with BCG at birth up to the first dose of measles administered at 9 months of age, as per EPI schedule, except for rotavirus vaccine (see Table 1).

Like the PDHS, the TPVICS included background characteristics of the child's household, parents' level of educational attainment, and wealth. A household comprises either one person living alone or a group of people living at the same address, who may or may not be related, with common housekeeping, who either share at least one meal a day or share common living accommodation (that is, a living room or sitting room).⁷ Parental education level is classified into four categories: none (not attended formal schooling), primary education (1–5 years of formal education), middle (6–8 years of formal education), secondary (9–10 years), and higher (11 years or more). Households are divided into five categories (poorest, poor, middle, rich, and richest) of equal size (that is, quintiles each with 20% of the population) using the number and types of consumer goods they own, ranging from a television to a bicycle or car, and housing characteristics such as source of drinking water, toilet facilities, and flooring materials.^{8,9} For the purpose of the TPVICS, KP province included 25 districts and not the newly merged districts that were part of the Federally Administered Tribal Areas. The districts included for KP are: Abbotabad, Bannu, Batagram, Buner, Charsadda, Chitral, Dera Ismail Khan, Hangu, Haripur, Karak, Kohat, Kohistan, Lakki Marwat, Lower Dir, Malakand protected area, Mansehra, Mardan, Nowshera, Peshawar, Shangla, Swabi, Swat, Tank, Torghar, and Upper Dir.

⁷ OECD. 2013. "Household definitions in other statistical standards", in OECD Guidelines for Micro Statistics on Household Wealth, OECD Publishing, Paris.
<https://doi.org/10.1787/9789264194878-18-en>

⁸ Filmer, Deon, and Lant H. Pritchett. 2001. "Estimating Wealth Effects without Expenditure Data—or Tears: An Application to Educational Enrollments in States of India." *Demography* 115–132

⁹ PDHS. 2018.

Table 1: List of vaccines administered to a fully immunized child in Pakistan

Disease	Vaccine	Doses	Administered at (child's age)
Childhood tuberculosis	Bacille Calmette-Guérin (BCG)	1	Soon after birth
Poliomyelitis	Oral Polio Vaccine (OPV)	4	OPV0: soon after birth OPV1: six weeks OPV2: 10 weeks OPV3: 14 weeks
	Inactivated Polio Vaccine (IPV)	1	IPV1: 14 weeks
Diphtheria	Pentavalent vaccine (DTP + Hep B + Hib)	3	Penta1: 6 weeks Penta2: 10 weeks Penta3: 14 weeks
Tetanus			
Pertusis (Whooping cough)			
Hepatitis B			
Hib pneumonia and meningitis			
Measles	Measles	1	Measles1: 9 months

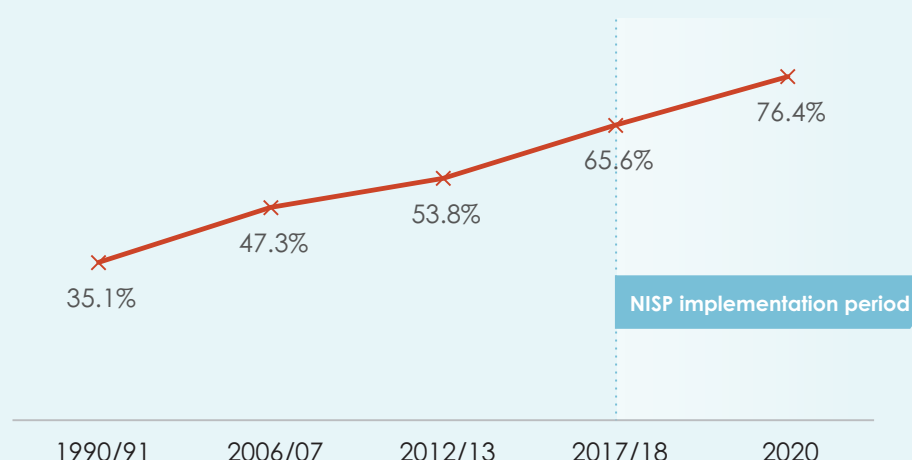
The report also uses data from the World Bank Databank (<https://databank.worldbank.org/home>) and the World Health Organization (WHO) immunization dashboard (<https://immunizationdata.who.int/>).

2. Immunization coverage in Pakistan: 1990 to 2020



Over the course of 30 years from 1990 to 2020, Pakistan more than doubled its immunization coverage (an increase of 117.6%). During that period the proportion of fully immunized children aged 12–23 months increased from 35% to 76% (see Figure 1). There was no noteworthy variation by gender in 2020, with 76.6% of female children and 76.2% of male children fully vaccinated. Similarly, there was no rural–urban divide, with 76.2% of children fully vaccinated in rural areas in 2020 compared to 76.7% in urban areas, for Pakistan overall. It is important to note that immunization service delivery in 2020 was hampered due to the outbreak of novel coronavirus disease (COVID-19) and, therefore, the immunization coverage rates were lower than planned.

Figure 1: Proportion of fully immunized children in Pakistan, 1990 to 2020



Source: TPVICS for 2020 data and PDHS for all other years

Greater immunization coverage over the period meant that Pakistan was able to reduce the incidence of two VPDs—diphtheria and measles—by 94% and measles by 88% respectively between 1990 and 2020 (see Figures 2 and 3). Diphtheria is a serious infection that causes difficulty in breathing, heart failure, paralysis, and even death¹⁰ especially among children.¹¹ Measles is a highly contagious disease and can be fatal for children under five years of age, the elderly, pregnant women, and those with a weakened immune system.¹² Both these diseases can be prevented through vaccination.

Pakistan also reduced the incidence of poliomyelitis (a disabling, sometimes fatal disease) between 1990 and 2020 from 777 to 135 cases. Despite this progress, Pakistan is one of the two countries in the world where poliomyelitis is still endemic.¹³ According to WHO, eradication of poliomyelitis requires high immunization coverage to block transmission of the virus that is extremely contagious. Several factors explain why some children are still not being vaccinated. These include lack of infrastructure, remote locations, population movement, conflict and insecurity, and resistance to vaccination.¹⁴

¹⁰ <https://www.cdc.gov/diphtheria/index.html>

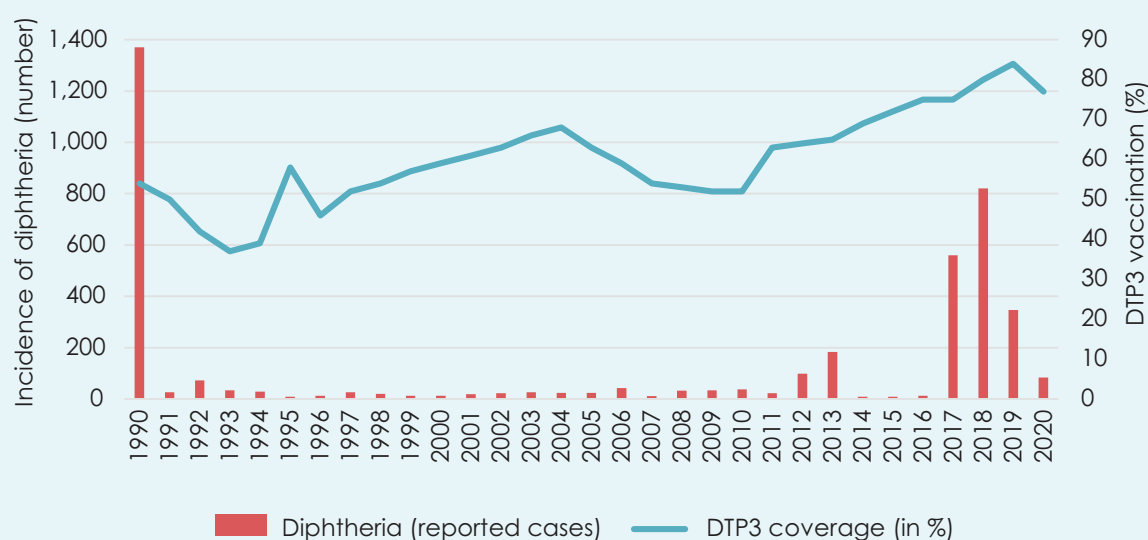
¹¹ <https://www.mayoclinic.org/diseases-conditions/diphtheria/symptoms-causes/syc-20351897#:~:text=Diphtheria%20can%20be%20treated%20with,be%20deadly%2C%20especially%20in%20children.>

¹² <https://www.cdc.gov/measles/symptoms/complications.html>

¹³ Global Polio Eradication Initiative, <https://polioeradication.org/where-we-work/polio-endemic-countries/>

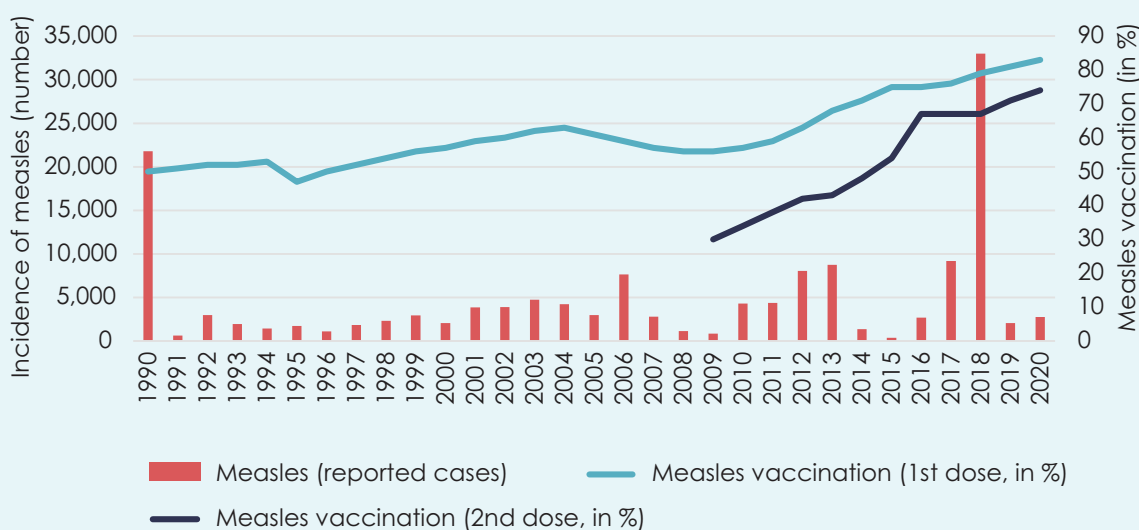
¹⁴ <https://www.who.int/news-room/feature-stories/detail/pakistan-and-afghanistan-the-final-wild-poliovirus-bastion>

Figure 2: Incidence of diphtheria and coverage of DTP3 vaccination: 1990 to 2020



Source: Data from WHO–UNICEF Joint Reporting available at: <https://immunizationdata.who.int>

Figure 3: Incidence of measles and coverage of measles first and second dose vaccination: 1990 to 2020



Source: Data from WHO–UNICEF Joint Reporting

Notes: Second dose of the measles vaccine was introduced in Pakistan in 2009;¹⁵ there was a measles epidemic in 2018 in Pakistan and supplementary immunization activities were conducted to vaccinate children against measles that year.¹⁶

¹⁵ <https://www.cdc.gov/mmwr/volumes/68/wr/mm6822a4.htm#:~:text=in%20all%20districts.,Immunization%20Activities,ages%209%20and%2015%20months.>
¹⁶ <https://www.unicef.org/pakistan/press-releases/unicef-and-who-join-forces-support-pakistans-mass-measles-immunisation-campaign>

Immunization coverage in the provinces

In 2020 the proportion of fully immunized children varied among the four provinces, from 38% in Balochistan to almost 90% in Punjab (see Figure 4). Nevertheless, the gains documented by each province since 1990 remain impressive (see Figure 5). Notable challenges to vaccine service delivery in Balochistan are difficult terrain and security issues. Interestingly, although KP also faces security issues, the vaccination coverage there is better than in Sindh, which has the country's second-largest provincial economy and better infrastructure than KP.¹⁷

Figure 4: Percentage of fully immunized children (aged 12 to 23 months) in Pakistan: 2020

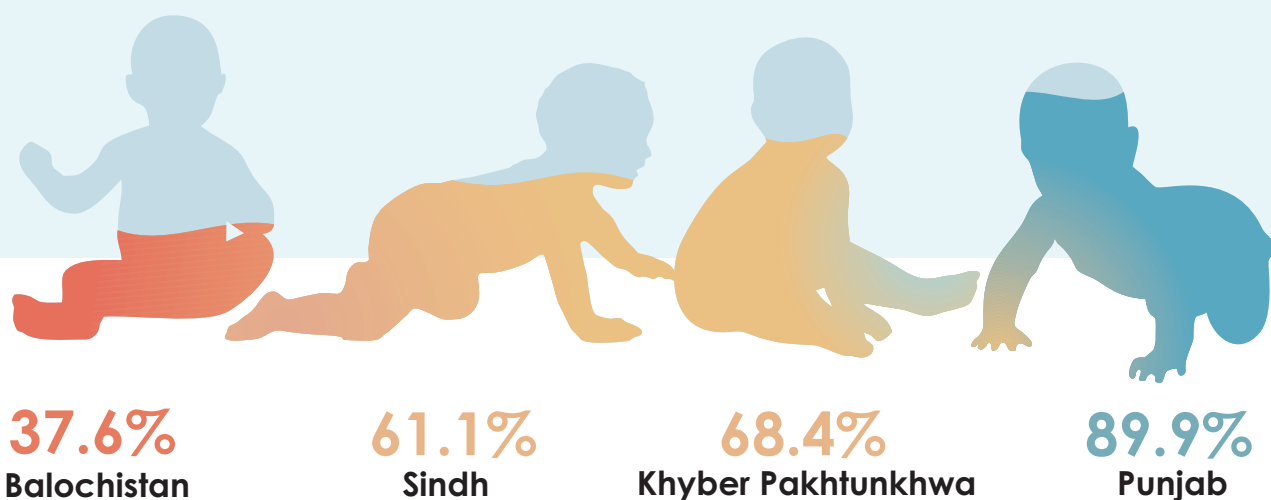
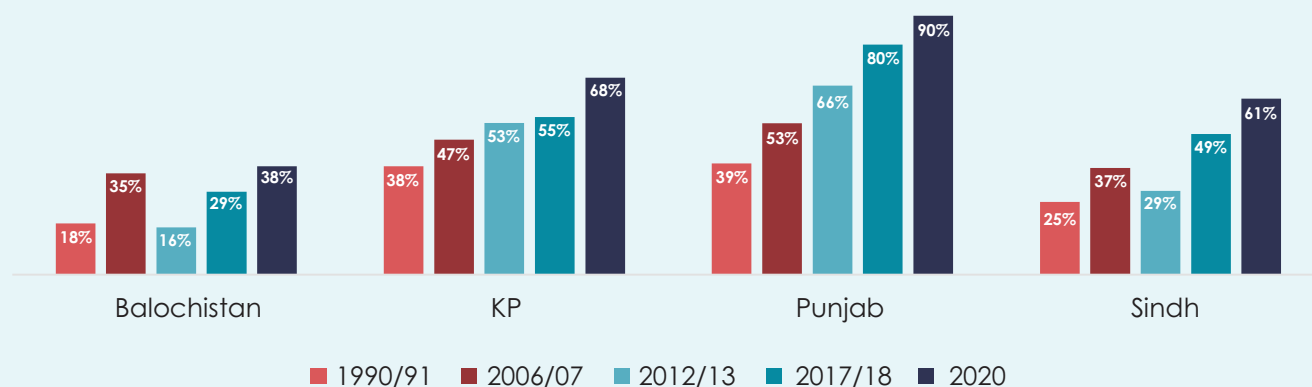


Figure 5: Percentage of fully immunized children (aged 12–23 months) in the provinces: 1990 to 2020

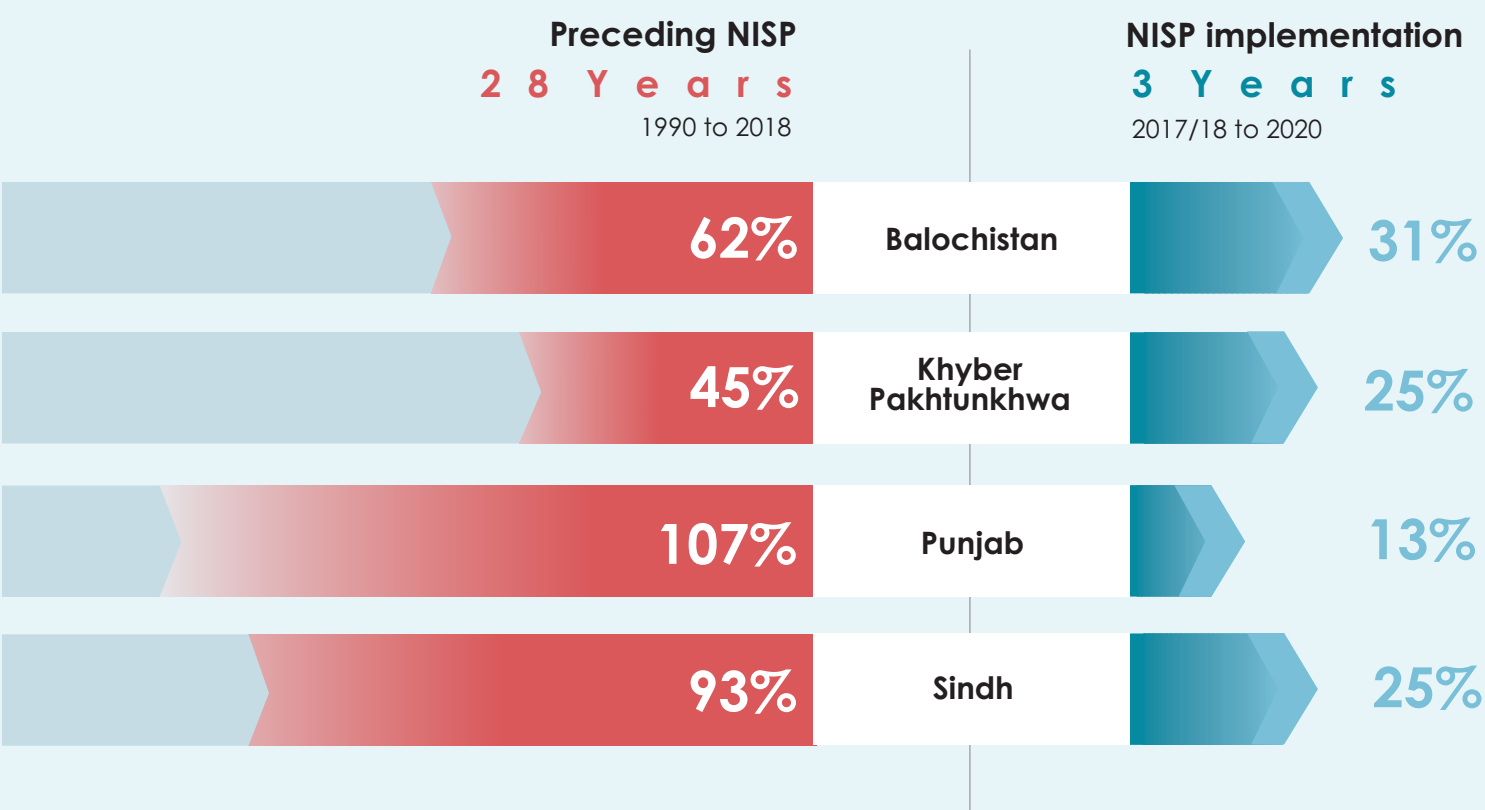


Source: TPVICS for 2020 data and PDHS for all other years

¹⁷ <https://en.wikipedia.org/wiki/Sindh>

KP and Punjab had the highest coverage in 2020, with 68% and 90% of fully immunized children respectively, while coverage in Sindh was 61% and in Balochistan 38% (the lowest in Pakistan). Nonetheless, this still represents an impressive increase of 20 percentage points or 111% for Balochistan since 1990. During the NISP implementation period, which started in 2017/18, all provinces documented significant improvements in immunization coverage. Over the 28 years preceding NISP, i.e., between 1990 and 2018, Balochistan increased the proportion of fully immunized children by 62%, KP by 45%, Punjab by 107%, and Sindh by 93%. By contrast, during the NISP implementation, in only three years between 2017/18 and 2020, Balochistan improved coverage by 31%, KP by 25%, Punjab by 13%, and Sindh by 25%. This points to the huge contribution made by NISP. Despite the progress summarized in this section, a relatively large proportion of children remain unvaccinated. In Balochistan, in 2020, one in every three children aged 12–23 months was not vaccinated; compared to roughly one in 10 in KP (11.4%) or Sindh (7.2%); whereas in Punjab only 0.7% of children were not vaccinated in 2020.

Figure 6: Improvements in percentage of fully immunized children (aged 12–23 months) during NISP period



Source: TPVICS for 2020 data and PDHS for all other years

Gaps in immunization coverage

Although immunization coverage has improved over time, disparities exist in terms of coverage by vaccine type, socioeconomic background, and demographics. There is considerable variation in coverage of each type of vaccine among the provinces. Punjab documented the highest coverage for every vaccine while Balochistan has lagged behind (see Table 2). Sindh lags considerably behind Punjab; for some vaccines, such as against polio and measles, the immunization rate in Sindh is lower than in KP.

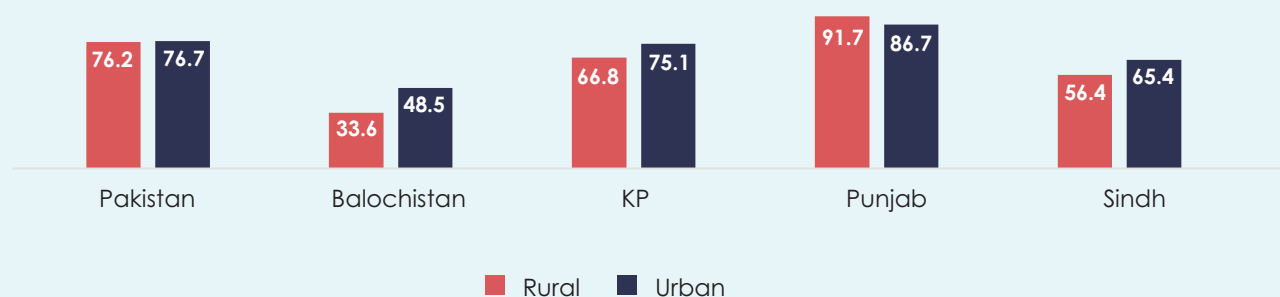
Table 2: Vaccination coverage by antigen: 2020

Disease	% of children age 12-23 months vaccinated by	Pakistan	Balochistan	KP	Punjab	Sindh
Childhood tuberculosis	BCG soon after birth	93.3	59.7	87.1	99.0	91.9
Poliomyelitis	OPV at birth (OPV 0)	92.5	58.0	85.9	98.5	90.9
	OPV 1 at 6 weeks	91.3	55.5	84.8	98.3	87.7
	OPV2 at 10 weeks	88.0	50.6	80.2	97.2	81.2
	OPV3 at 14 weeks	84.1	45.1	75.2	94.9	74.9
	Inactivated polio vaccine (IPV) at 14 weeks	84.2	44.7	75.9	95.4	73.7
DTP, Hep B, Hib	Pentavalent vaccine (Penta 1) at 6 weeks	90.4	52.3	83.2	98.2	85.9
	Penta 2 at 10 weeks	87.4	47.6	79.6	97.1	79.6
	Penta 3 at 14 weeks	83.5	42.2	74.3	94.9	73.2
Measles	Measles	80.4	42.6	73.1	92.5	67.1
	Measles 2 (15 to 23 months)	55.7	12.8	49.5	69.1	41.3

Source: TPVICS 2021

Besides, there is a wide variation in immunization coverage between rural and urban areas with the rural areas lagging significantly in Balochistan, KP and Sindh. In Punjab, there were proportionally more fully immunized children in rural areas than in urban areas (see figure 7). Focusing exclusively on rural areas, the data show that roughly one-third of the children were vaccinated in Balochistan, a little more than half in Sindh, and 67% in KP. This clearly indicates the need for ramping up efforts to vaccinate children in the rural areas.

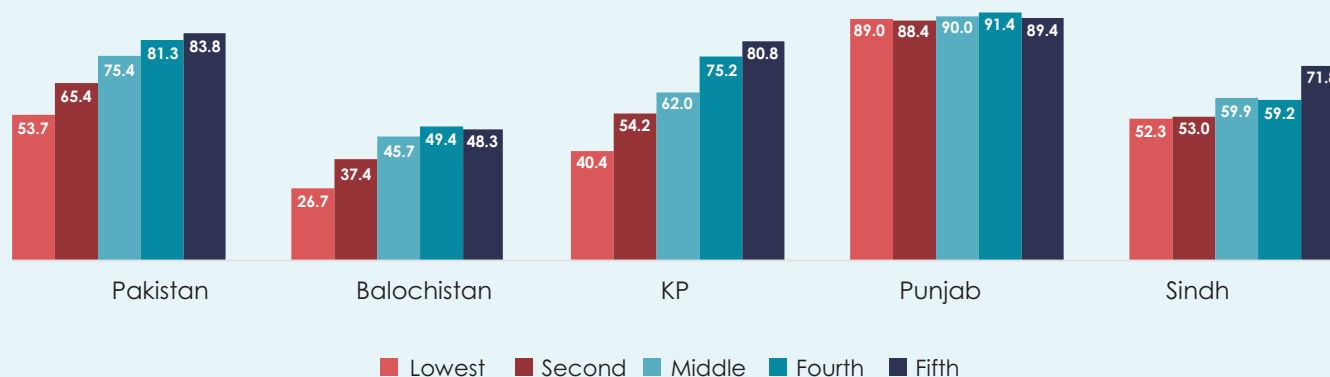
Figure 7: Percentage of fully immunized children (aged 12–23 months) by residence: 2020



Source: TPVICS 2021

As expected, a household's wealth has a bearing on the likelihood of its children being vaccinated. The two lowest income quintiles lag significantly compared to the richer quintile for all provinces, except Punjab, where coverage is comparable across the income groups (see Figure 8). For Sindh, the difference between the richest quintile and the four lower income quintiles is stark—immunization coverage among the richest was 20% more than in the middle-income quintile and 37% greater than in the poorest quintile. Intriguingly, the immunization coverage among the richest quintile in Sindh was lower than in KP. For Balochistan, the three richest quintiles had similar immunization coverage levels.

Figure 8: Percentage of fully immunized children (aged 12–23 months) by household wealth quintile: 2020



Source: TPVICS 2021

Factors that influence vaccination uptake

The parents' education plays a key role in the vaccination status of a child. In Balochistan and Sindh the father's education played a stronger role (than the mother's) whereas in KP and Punjab, mother's education played a stronger role than father's education (see Table 3).

Table 3: Percentage of fully immunized children (aged 12–23 months) by parent's educational level: 2020

Education level	Pakistan	Balochistan	KP	Punjab	Sindh
Mother's education:					
No education	67.9	35.5	60.5	89.1	53.3
Primary	83.3	27.5	76.3	90.6	63.2
Middle	81.3	36.7	78.5	88.4	62.4
Secondary	83.8	47.8	79.3	90.6	71.7
Higher	87.1	66.2	82.4	91.1	81.5
Father's education:					
No education	88.9	51.1	58.3	34.5	66.8
Primary	89.1	56.9	66.6	37.3	78.8
Middle	90.4	64.1	70.0	35.3	82.7
Secondary	90.8	68.3	74.4	43.8	82.6
Higher	90.7	76.1	79.1	63.5	83.6

Source: TPVICS 2021

Infrastructural issues including proximity to the nearest vaccination site, mistrust in the overall vaccination program, and lack of information were some of the reasons for vaccine hesitancy in 2020. The table below indicates the reasons stated by parents of children who did not receive all the required vaccinations. This points to the need for strengthened information campaigns alongside enhanced service delivery systems.

Table 4: Reasons for vaccine hesitancy (expressed) as a percent of children not fully vaccinated): 2020

Reasons for vaccine hesitancy	Balochistan	KP	Punjab	Sindh
Infrastructure/program related issues:				
Place of immunization too far	6.5	1.7	0.1	1.4
Time of immunization not convenient	1.3	0.4	0.0	0.2
Took child to the site but no vaccine was available	0.3	0.0	0.0	0.1
Took child to the site but no vaccinator was there	0.5	0.1	0.0	0.0
Long wait	0.9	0.2	0.0	0.2
Information gap:				
Rumors	8.6	3.9	0.2	1.6
No faith in immunization	5.2	3.6	0.2	2.6
Fear of reaction (side effects)	3.1	1.1	0.1	1.0
Time or place of immunization not known	3.6	0.5	0.0	0.4
Took child to the site but it was closed	0.3	0.1	0.0	0.0
Took child to the site but not on the vaccination day	0.3	0.1	0.0	0.0
Other:				
Mother too busy	2.9	0.7	0.1	0.7
Family problem including mother ill	1.5	1.1	0.0	0.4
Child ill, not taken to the vaccination site	1.8	1.0	0.0	0.7

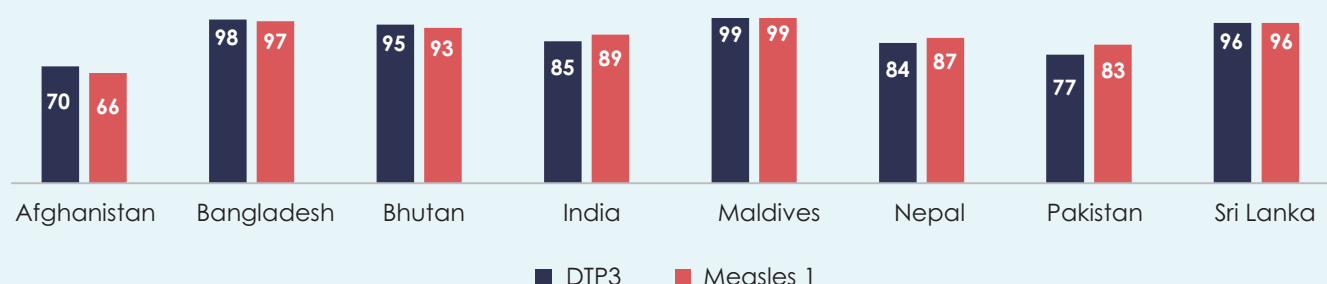
Source: TPVICS 2021

3. A Regional comparison



This section provides an overview of where Pakistan stands in immunizing its children with respect to other countries in the region. With respect to coverage of DTP3 and measles (first dose) vaccination, Pakistan had the second lowest coverage in the South Asia region in 2020 (see Figure 9).

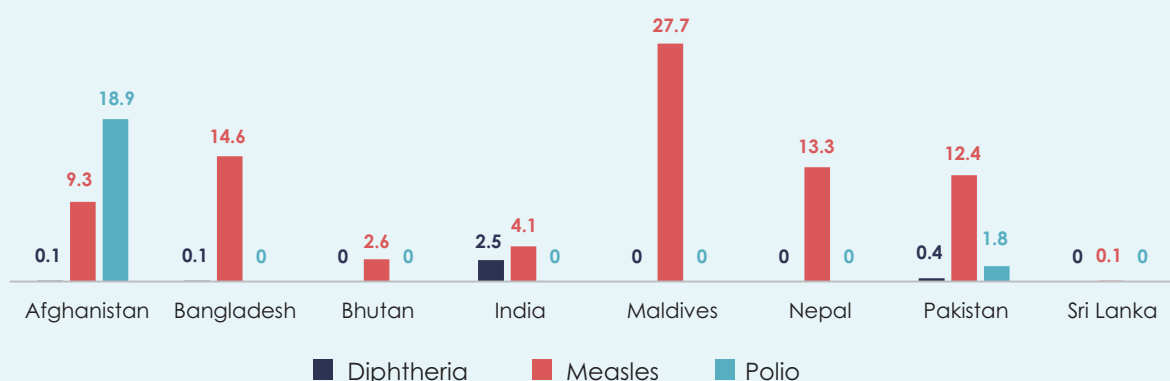
Figure 9: Coverage of DTP3 and measles first dose vaccination among children aged 12–23 months (%): 2020



Source: Data from WHO–UNICEF Joint Reporting

In terms of incidence of VPDs, Pakistan had the second highest incidence of both diphtheria and polio in South Asia in 2020. With respect to measles, the incidence in Pakistan was the fourth highest (see Figure 10). Pakistan and Afghanistan are the only two countries in the world where polio is still prevalent. All others including those in South Asia have all been declared polio free by WHO.

Figure 10: Incidence of selected VPDs in South Asia: 2020

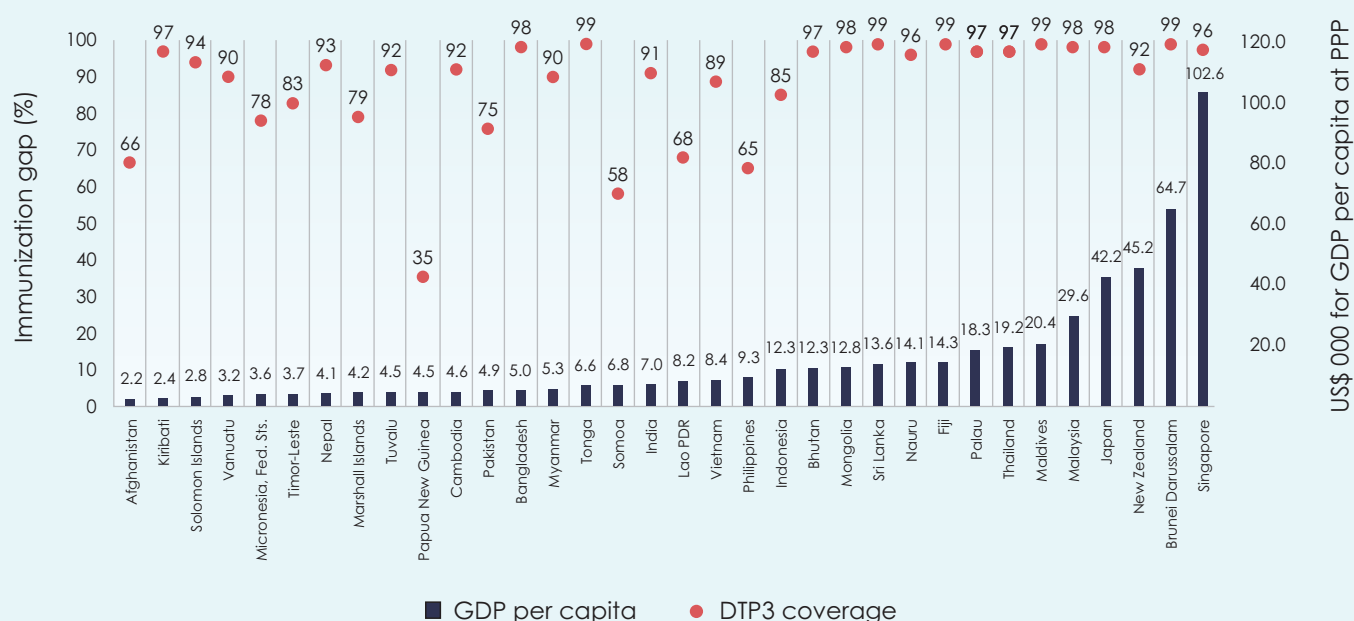


Source: Data from WHO–UNICEF Joint Reporting

Note: the figures are expressed as cases per 1 million population. Incidence rate of diphtheria for Afghanistan is for 2019; measles for Afghanistan and Bhutan for 2019.

In comparison to the countries in Eastern and Southern Asia that have similar income levels to those of Pakistan, immunization coverage is relatively low (see Figure 11). Countries of comparable wealth, such as Cambodia, Nepal, Bangladesh, or Myanmar had much higher coverage of DTP3 vaccines in 2019. Pakistan with its GDP per capita of US\$4,889 had a DTP3 coverage rate of 75% compared to Cambodia with GDP per capita of US\$4,574 and 92% DTP3 coverage, Nepal US\$4,120 and 93% coverage, Bangladesh US\$4,955 and 98%, or Myanmar US\$5,297 and 90%. This points to the need for Pakistan to further strengthen its immunization program to improve the health of its children. In addition, there is a need to further improve efficiency of the EPI program. As reflected in Figure 12, health expenditure per capita in Pakistan is mostly at par with other countries in South Asia. The immunization coverage rate, however, is comparatively low.

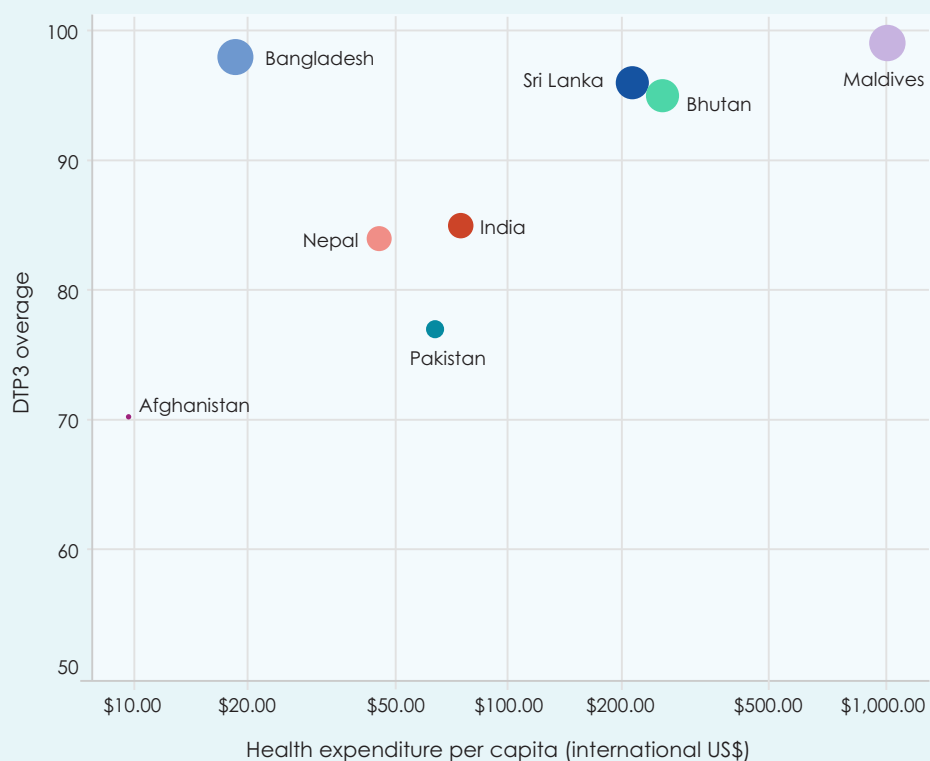
Figure 11: A comparison of child vaccination status (DTP3) with respect to wealth in Asia: 2019



Source: World Bank Databank

Notes: 2019 data used for this figure. Per capita values for GDP are expressed in current international dollars converted by purchasing power parity (PPP) conversion factor. DTP3 coverage means three doses of DTP vaccines administered to children aged 12–23 months. Countries included in this figure are drawn from the World Bank's South Asia and East Asia and the Pacific regions.

Figure 12: Comparison of health expenditure and immunization coverage in South Asia



Source: World Bank Databank

Notes: Domestic health expenditure per capita expressed in current international dollars converted by PPP conversion factor. Log of health expenditure is used in this graph. DTP3 coverage means three doses of DTP vaccines administered to children aged 12–23 months. For health expenditure, 2018 data has been used, while for DTP3 coverage, 2020 data has been used. Larger dots indicate higher DTP3 coverage



4. Conclusions and way forward

The previous section documented Pakistan's impressive progress in immunization coverage (a 117.6% increase in fully immunized children between 1990 and 2020). Consequently, the burden of two VPDs declined, diphtheria by 94% and measles by 88%, over the same period. Despite these improvements, wide variations remain in terms of coverage by vaccine type, socioeconomic background, demographics, as well as between and among provinces. More importantly, Pakistan's immunization coverage is low compared to other countries in the region. The burden of disease remains high, with the incidence of both diphtheria and polio in Pakistan being the second highest in 2020. More worrying still is that Pakistan is one of only two countries in the world where polio is still prevalent. When compared to countries in Eastern and Southern Asia that have a similar level of wealth, Pakistan's immunization coverage ranks very low. These facts point to the need to strengthen Pakistan's immunization program to improve the health of its children.

- **Prioritize lagging areas:** Concerted efforts are needed to maximize the coverage in underperforming provinces and districts by increasing focus on supervision, as well as monitoring and evaluation. To target lagging areas, district-specific strategies can be developed that are better aligned with local norms and culture. Performance-based incentives for outreach workers and vaccinators could be explored to improve vaccine uptake. For hard-to-reach areas, mobile teams can be created.
- **Further strengthen service delivery systems:** The number of vaccination sites may need to be increased to reduce waiting time, particularly in densely populated areas. The capacity of the district-based EPI teams needs to be strengthened to improve service delivery. Training and use of information technology can enhance the capacity of the vaccinators and EPI teams.
- **Ensuring sustainable financing for immunization:** One of the key achievements under NISP is the transfer of immunization-related budget from development partner financing to the government's share. In addition, dedicated accounting codes have been created to track immunization-related budget and expenditure. This helps the government in projecting total requirements and estimating needs. Given the changing burden of diseases, it is critical to ensure that the budgetary needs for immunization are adequately financed and the program is not hampered due to financial constraints.
- **Foster partnership:** Partnership with the private sector can improve the efficacy of the EPI program. It can also help in promoting vaccine uptake and reaching out to areas where there are limited vaccination sites. There are good practices documented by other countries in the region that Pakistan can explore.
- **Explore innovative approaches for increasing demand for vaccination:** conditional or co-responsibility cash transfers (CCT) have proven to be effective in improving HNP outcomes, particularly creating demand for immunization, in many countries in Asia and Africa. Under a CCT, cash is given to targeted beneficiaries upon fulfillment of conditions or coresponsibilities designed to promote uptake of selected services, such as vaccination. This is being piloted under NISP and can be rolled out if proven successful in the context of Pakistan.

